

Shadows Left Behind

<https://github.com/kcarrubba/CMS320-Fall2025>

CMS 320: Video Game Design

Final Project Report

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Introduction

Game Overview

Shadows Left Behind is a 2.5D single-player story game set within a quiet, smoke-stained house filled with fragments of memory. The player moves through rooms once touched by warmth and now hollowed by loss, collecting glowing clues that reveal what happened on the night of the fire, a tragedy forever bound to the memory of the protagonist's mother.

Each room ends with a Quest, a symbolic challenge between Light and Darkness. Choosing Light or Darkness doesn't change difficulty, only tone. If the player chooses Light and succeeds, the next scene appears in Light; if they fail, it shifts into Darkness — and vice versa. The game explores not winning or losing, but how emotion reshapes the world around the player.

This idea was chosen for its ability to create a strong emotional atmosphere using simple, accessible mechanics. The story is told entirely through visuals, sound, and contrast rather than dialogue, making it immersive and expressive for a small creative team.

Target Users

The game is designed for players aged 12 and older who enjoy atmospheric, reflective, and story-driven experiences.

- Fans of indie titles such as Sally Face, Inside, and Gris, where tone and environment carry the emotion.
- Players who appreciate calm yet meaningful games that blend art, music, and subtle storytelling into a cohesive emotional experience.

Game World

The entire story unfolds inside a memory-wrapped house, suspended between the past and the present. Each room, the Bedroom, Hallway, Kitchen, and Attic, reveals fragments of a night that once glowed with life and ended in silence.

The house exists in two shifting states:

- Light Realm: serene, golden light through cracked windows, soft melodies, and an atmosphere of gentle acceptance.
- Darkness Realm: cool, ashen tones with faint embers and quiet echoes where warmth once lived.

The world doesn't become harder or easier, it simply changes in feeling. Light and Darkness are not enemies but emotional reflections of grief and acceptance, both necessary to complete the story.

Rules

- Explore each room and collect three glowing clues to unlock the Quest.
- After collecting all clues, choose between Light or Darkness to begin the Quest.
- Complete the short Quest:
 - If you choose Light and succeed, the next scene appears in Light.
 - If you choose Light and fail, the next scene appears in Darkness.
 - If you choose Darkness and succeed, the next scene appears in Darkness.
 - If you choose Darkness and fail, the next scene appears in Light.
- The choice and outcome affect only the tone of the next room, not its difficulty.
- The final room reflects the path the player has taken, concluding the story through light, shadow, or a fragile mix of both.

Controls

- A / W / S / D is Move
- E is Collect clue or interact
- Space is Jump in one of the mini-games
- Mouse Click is Interact in mini-games

(Keyboard-based gameplay for simplicity and accessibility.)

Task Distribution Among Teammates

- **Ethan, Lead Programmer**
Responsible for building the game's core technical foundation. He develops interaction logic, Quest mechanics, and Light/Darkness transitions. He ensures the game runs smoothly, implements system-level scripts, and integrates all content into a cohesive, stable experience. He also fully develops his own mini-game.
- **Kaley, Lead Artist, Mini-Game Programmer**
Defines the visual identity of the game through detailed pixel-art environments, player movement, clue illustrations, UI elements, and lighting variations for both Light and Darkness realms. She also fully develops her own mini-game and assists with other programming needs.
- **Katerina, Narrative Designer, Audio Designer, Mini-Game Programmer**
Leads the narrative vision of the project, shaping story progression and meaning behind each clue and room. She oversees player experience design, ensuring that tone, visuals, mechanics, and symbolism work seamlessly together. In addition, she develops all audio elements like ambient soundscapes, emotional cues, and mini-game effects, and fully develops her own mini-game.

Game Development

Ethan

Core Architecture and Scene Flow

- Built the early game framework by adding the GameManager and the core methods to switch scenes.
- Added foundational scenes including the Main Menu and the Bedroom, and standardized a global EventSystem approach to reduce duplicated UI systems across scenes.

Player System

- Created the player prefab and implemented the PlayerController to enable consistent character behavior across scenes.
- Implemented and refined player clamping/bounds logic (including collider-based approaches and later adjustments tied to sprite changes), helping ensure the player stays correctly constrained within visible/interactive space.
- Addressed early startup issues with fixes targeting scene initialization bugs.

Spawnpoint System and Persistent Player Handling

- Added a spawnpoint prefab and integrated spawnpoint logic into scene transitions.
- Reworked the flow so the player prefab is managed globally through the GameManager, improving consistency and preventing duplicate player instances across scene loads.
- Continued to refine spawn behavior with later commits that adjust backgrounds/spawn points as art and level structure evolved.

Interactables Framework

- Added the interactable prefab and the supporting scripts for interactable entities, laying down the system for clue-driven progression.
- Implemented an InteractablesManager to track whether interactions have been completed and ensured the game only counts a clue as discovered after the popup is properly closed.

Overlay and UI

- Improved the overlay system by:
 - Moving overlay components into a CanvasGroup structure for reliable fading behavior.
 - Adding and fade behaviors for interactables and the intro.
 - Fixing the issue where the first popup would not fade in correctly.
 - Allowing popups to be closed by pressing the interaction key again.

Doors and Progression

- Replaced earlier scene-switching UI with a more immersive approach by adding a Door prefab and related logic.
- Implemented gating so doors only appear when all required clues are interacted with.
- Added doors into multiple scenes and refined door behavior to work reliably.

Light/Dark State Integration

- Contributed to the game's light and dark room system by:
 - Making room backgrounds into a prefab-based approach for easier swapping.
 - Implementing logic to set up rooms based on game outcome and ensuring minigame or puzzle states correctly signal the light/dark flag.

Minigame Implementation and Fixes

- Built and integrated parts of the third minigame, including:
 - Creating a dedicated minigame scene and implementing core minigame logic.

- Adding a timer and ensuring win/loss transitions lead to the correct single next scene rather than branching inconsistently.

Kaley

Project Initialization & Documentation

- Created the Unity project and initial repository structure.
- Wrote and updated the project's README to outline goals and setup instructions.

Visual Design & Scene Art

- Added and refined multiple scene backgrounds, including:
 - Bedroom background
 - Light and dark versions of hallway, kitchen, and attic backgrounds
 - Hallway jumper, puzzle, and fire picking game backgrounds and elements
- Resolved sprite quality issues and scaling problems to preserve crisp pixel-art style.

Puzzle Game Creation

- Fully created the entire puzzle minigame, including layout, functionality, and scene setup.
- Added the puzzle scene and integrated its visuals and mechanics into the broader game flow.

Minigame & Gameplay Enhancements

- Designed obstacles for the infinite hallway runner game.
- Added animation assets, fixed animation issues, and updated animation metadata.
- Added fire animation to prefab and coordinated how it plays in the fire minigame.

Scene Interaction & Door System

- Designed and added the initial door object and updated its script logic.

Lighting / Dark-Mode System Contributions

- Added light and dark variants for multiple rooms (hallway, kitchen, attic).

UI & Fonts

- Updated and integrated custom fonts (PressStart2P) into the UI.
- Updated the fire and puzzle minigame timer fonts and other text elements.

Bug Fixes & Polishing

- Fixed issues involving animation playback.
- Fixed issues involving executable file formatting and created executable.

Katerina

Narrative Design & Story Development

- Wrote the complete narrative foundation for Shadows Left Behind, including the emotional arc, room progression, and overall thematic tone.
- Created the detailed clue system and all clue descriptions (12 total), ensuring every object fit the same emotional language, worldbuilding style, and narrative symbolism.
- Designed the ending letter and all in-game written elements to unify tone, pacing, and emotional clarity across the project.

Mini-Game Programming & Gameplay Flow

- Programmed the Hallway Runner mini-game using a structure inspired by Google Dino Runner, then expanded it with custom mechanics, new prefab types, and unique obstacle behavior.
- Coordinated obstacle integration with Kaley's art to ensure consistent difficulty and visual coherence.
- Participated in resolving game-flow issues, refining scene transitions, and improving overall pacing based on playtesting feedback.

Audio Design & Implementation Planning

- Selected, curated, and prepared all background music for the game, including distinct ambient tracks for every room and each mini-game.
- Planned audio behavior for interactive scenes to ensure a cohesive sound experience throughout the game.

Visual Asset Contributions

- Created all close-up clue images (approximately 12 total), including lighting style, mood, and pixel-art consistency.
- Added narrative captions directly into each clue design to ensure visual and emotional alignment with the rest of the game.
- Assisted with sprite importing, scaling, and optimizing image clarity to maintain consistent pixel-art presentation across scenes.

Project Coordination & Team Support

- Organized team tasks and helped maintain communication flow between programming, art, and design roles.
- Participated in playtesting and debugging sessions, identifying issues with pacing, clarity, or scene logic.
- Contributed to solving narrative coherence problems and aligning game mechanics with story goals.

Final Game

https://youtu.be/aw7_b5hw6s

Discussion and Future Work

Developing *Shadows Left Behind* came with several early challenges, especially when trying to define the flow of the game. In the beginning, we struggled with how rooms should connect, how clues should appear, and how the player would naturally progress through the house. The project originally had a completely different concept: two prefabs of the same character on screen at once, one representing Light and the other Darkness. While visually interesting, we quickly realized that players wouldn't have a meaningful incentive to control two characters at the same time. It felt confusing, overwhelming, and didn't support the emotional tone we wanted. This led us to simplify the design and shift our focus toward storytelling, atmosphere, and smoother gameplay.

Playtesting helped guide these decisions even further. Players mentioned they needed a clearer introduction to understand why they were in the house and what clues represented, which pushed us to strengthen the opening narrative. Another major challenge was creating smooth and responsive controls, especially after early playtests of our mini-game showed frustration with movement glitches. By adjusting physics settings, refining scripts, and testing repeatedly, we were able to deliver a much more stable and enjoyable interaction system. We also dedicated time to building a consistent art and audio style, ensuring every clue felt connected emotionally and visually. These insights shaped our design priorities, clarity, emotional pacing, and creating challenges that are engaging without being frustrating.

Looking ahead, we see many opportunities to expand *Shadows Left Behind*. We would like to enlarge each room to allow for deeper exploration and add more clues, including hidden or optional ones, to let players uncover additional layers of the story at their own pace. More complex or varied mini-games could strengthen the Light/Darkness mechanic, offering greater interactivity without losing accessibility. Additional scenes, subtle animations, and expanded sound design could heighten emotional impact, while further accessibility options would help reach a wider audience. Overall, this project laid a strong foundation, and we are excited by the potential to grow the game into a richer, more immersive narrative experience.